



SESHADRIPURAM EDUCATIONAL TRUST
SESHADRIPURAM COLLEGE
POST GRADUATE DEPARTMENT OF COMPUTER APPLICATIONS
#27, Nagappa Street, Seshadripuram, Bengaluru-560020
AFFILIATED TO Dr.MANMOHAN SINGH BENGALURU CITY UNIVERSITY |
AICTE APPROVED
ISO 9001:2015 CERTIFIED | **NAAC GRADE A++**
department.mca@spmcollege.ac.in | <https://www.spmcollege.ac.in/mca/>

PROJECT REPORT ON
**“SHYRA - SMART WOMEN SAFETY ALERT
SYSTEM”**

Submitted in partial fulfillment for the award of degree in
MASTER OF COMPUTER APPLICATIONS (MCA)

III SEMESTER

Submitted by,

Chandana M – P18GH24S126008

Chandana M – P18GH24S126073

Under the guidance of

Prof. Bharathi N

Academic Year: 2025-2026

CERTIFICATE

This is to certify that the mini project entitled “**SMART WOMEN SAFETY ALERT SYSTEM**”, carried out by **Chandana M (P18GH24S126008)** and **Chandana M (P18GH24S126073)**, has satisfactorily completed the course of experiments in the practical project for the award of Master of Computer Applications in the Post Graduate Department of Computer Applications, Seshadripuram College, Affiliated to **Dr.Manmohan Singh Bengaluru City University**, during the academic year of 2025-2026.

Project guide

Prof. Bharathi N

Incharge Head,

Prof.Rajeshwari V

Date:

Examiners

1.....

2.....

DECLARATION

We, **CHANDANA M - P18GH24S126008** and **CHANDANA M - P18GH24S126073**, hereby solemnly declare that the mini project entitled **“SMART WOMEN SAFETY ALERT SYSTEM”** is a bonafide work carried out by us under the guidance of **Prof. Bharathi N**, Post Graduate Department of Computer Applications, Seshadripuram College, Bengaluru.

This project has been submitted in partial fulfilment of the requirements for the degree of Master of Computer Applications (MCA) of Dr. Manmohan Singh Bengaluru City University.

Place: Bengaluru

Date:

Chandana M – P18GH24S126008

Chandana M – P18GH24S126073

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ABSTRACT

In recent years, personal safety has become a major concern, particularly for women who may face unsafe situations in daily life. With the rapid advancement of smartphone technology, mobile applications can play an important role in providing quick and reliable assistance during emergencies. The Shyra – Smart Women Safety Android Application is developed to provide a simple, efficient, and accessible safety solution that helps users quickly alert trusted contacts and share critical information when they feel threatened or unsafe.

The main objective of this project is to design and develop an Android-based application that enables women to send emergency alerts, share live location, and collect evidence during critical situations. The application integrates multiple safety features into a single platform so that users can easily access them with minimal interaction. The system is developed using the Java programming language with XML-based layouts for the user interface, following standard Android development practices. For backend services, the application uses Firebase Authentication for secure user login and Firebase Realtime Database for storing user information and emergency contacts.

One of the most important features of the application is the SOS emergency alert system. The app provides a large SOS button on the main dashboard that allows the user to instantly send an emergency message to their saved contacts. When the SOS button is pressed, the application retrieves the user's current location using the device's GPS through the Fused Location Provider API. The location is converted into a Google Maps link and automatically sent to all registered emergency contacts through SMS. This allows family members or friends to quickly identify the user's location and provide immediate assistance.

The application also includes emergency contact management, which allows users to add, edit, or delete important contacts such as family members or trusted friends. These contacts are stored securely in the Firebase database under the user's account. Additionally, the application provides GPS location sharing, allowing users to manually share their current location through messaging applications such as SMS or WhatsApp.

To enhance safety and provide potential evidence in emergency situations, the application also includes camera and audio recording features. Users can capture photos using the device's camera or record audio using the MediaRecorder API. These files can be shared immediately through installed applications, or stored locally on the device if sharing is not possible. The system prioritizes user privacy by keeping media files stored locally rather than uploading them to the cloud.

The user interface of the application is designed to be simple and user-friendly using Material Design components. Navigation is implemented using a navigation drawer and swipeable pages with ViewPager2, allowing users to easily access different sections such as the home dashboard, safety tips, and about page. The app also includes helpful safety tips that educate users about precautionary measures and personal safety practices.

In conclusion, the Shyra Women Safety Application provides a practical mobile-based safety solution that combines emergency alerts, location tracking, and evidence collection into a single platform. By utilizing modern smartphone capabilities and secure backend services

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CHAPTER-1

INTRODUCTION

INTRODUCTION

1.1 Introduction

In today's world, personal safety has become a significant concern, especially for women who may encounter unsafe situations in public places, workplaces, or while traveling alone. With the rapid advancement of mobile technology and the widespread use of smartphones, mobile applications have become powerful tools for improving personal security and enabling quick communication during emergencies. Android smartphones, equipped with features such as GPS, internet connectivity, camera, and messaging services, provide an excellent platform for developing safety-oriented applications.

The Shyra – Smart Women Safety Android Application is designed to provide a reliable and easy-to-use safety solution for women. The application integrates several emergency features into a single platform that allows users to quickly alert their trusted contacts in critical situations. The app enables users to send emergency SOS messages with live location, manage emergency contacts, capture photos or audio recordings, and trigger alerts using simple actions such as pressing a button or the device's power key.

The application is developed using Java programming language with XML layouts for the user interface, following Android development standards. For backend services, the application uses Firebase Authentication for secure user login and Firebase Realtime Database to store user-related data such as emergency contacts. By combining modern mobile technologies with user-friendly design, the Shyra application aims to provide a practical tool that enhances personal safety and enables faster emergency response.

1.2 Importance and Relevance to Real World Problems

Women's safety has become a major social issue in many parts of the world. Incidents of harassment, assault, and other forms of violence highlight the need for effective safety mechanisms that can help women quickly seek assistance during emergencies. In many situations, victims may not have enough time to call someone or explain their situation. Therefore, a fast and reliable alert system becomes extremely important.

Smartphones are widely used and always carried by individuals, making them ideal tools for implementing safety solutions. By utilizing smartphone features such as GPS location tracking, messaging services, camera, and audio recording, safety applications can help users communicate their situation instantly and provide useful information to trusted contacts.

The Shyra Women Safety Application addresses these real-world problems by providing features such as instant SOS alerts, live location sharing, emergency contact management, and evidence capture through camera and audio recording. These features allow users to quickly inform their family members or guardians and provide accurate location details, enabling faster help and support during critical situations.

1.3 Problem Statement

Despite advancements in technology, many women still face safety risks in daily life. In emergency situations, it may be difficult for a person to make a phone call, send messages, or provide their location to others quickly. Traditional methods of seeking help may take too much time or require multiple steps, which may not be possible during dangerous situations.

Additionally, many existing safety applications either lack essential features, require constant internet connectivity, or have complicated interfaces that are difficult to use during emergencies. There is a need for a simple, reliable, and efficient safety application that allows users to send emergency alerts quickly and share their location with trusted contacts.

Therefore, the main problem addressed in this project is the lack of a comprehensive and easy-to-use mobile application that can provide immediate assistance and communication during emergencies.

1.4 Limitations of the existing project

Although several women safety applications are available, they have certain limitations that reduce their effectiveness in real-world situations.

Many existing applications provide only basic SOS functionality without integrating additional features such as evidence collection, multiple alert methods, or discreet triggering mechanisms. This limits their usefulness during complex emergency situations.

Another major limitation is the dependence on internet connectivity. Some applications require continuous internet access to send alerts or share location data, which may not be available in all situations, especially in remote or low-network areas.

In addition, several applications have complex user interfaces, making them difficult to operate quickly during emergencies. In stressful situations, users require simple and easily accessible features, which many existing systems fail to provide.

Another drawback is the lack of discreet activation methods. Most applications require users to manually open the app and press an SOS button, which may not be possible in certain situations where the user is under threat.

Furthermore, some applications store sensitive data such as images and recordings in cloud storage, raising privacy and security concerns for users.

These limitations highlight the need for an improved system that provides a comprehensive set of safety features, ease of use, reduced dependency on internet connectivity, and better privacy protection. The Shyra Women Safety Application is designed to address these issues effectively.

1.5 Proposed Solution

To address the identified problem, the Shyra Women Safety Application is proposed as an Android-based safety system that provides multiple emergency assistance features in a single platform. The application allows users to register and securely log in using Firebase Authentication. Once logged in, users can add and manage emergency contacts who will receive alerts during emergencies.

The application provides an SOS feature that sends emergency SMS messages containing the user's live GPS location in the form of a Google Maps link. This ensures that emergency contacts can quickly identify the user's exact location. The application also includes a power button SOS trigger, which allows users to activate emergency alerts discreetly even when the phone is locked.

In addition, the application includes GPS location sharing, camera photo capture, and audio recording features that allow users to collect and share important information during emergencies. The application interface is designed using Material Design principles to ensure ease of use and quick navigation between different sections such as the home screen, safety tips, and about page.

1.6 Scope of the Project

The scope of the Shyra Women Safety Application is focused on developing a mobile-based safety system that provides essential emergency support features for users. The application primarily targets women who may require quick access to emergency assistance in unsafe situations.

The system allows users to create an account, store emergency contacts, send SOS alerts with live location, share GPS location, capture photos, and record audio during emergencies. The application also provides safety tips to increase awareness about personal safety practices.

The project focuses on Android devices and uses Firebase services for authentication and database management. The system is designed to work on smartphones running Android API level 21 and above. The application aims to provide a reliable safety tool that can be easily installed and used by individuals without requiring advanced technical knowledge.

CHAPTER-2

LITERATURE SURVEY

LITERATURE SURVEY

2.1 Reviewed Literature and Technologies

Women's safety has become an important topic in recent years, and many researchers and developers have proposed different technological solutions to address this issue. Several mobile applications and research studies have focused on using smartphone capabilities such as GPS tracking, messaging systems, sensors, and internet connectivity to provide emergency support during unsafe situations.

One commonly studied solution is the use of GPS-based tracking systems in safety applications. These systems allow users to share their real-time location with trusted contacts or emergency services. GPS technology enables accurate tracking of a user's position and helps responders reach the location quickly. Many existing safety applications use GPS combined with messaging services to send location links during emergencies.

Another widely used approach in women safety applications is the SOS alert system. In many applications, a single button is used to trigger an emergency alert that sends messages to predefined contacts. Some applications also include automated calling features that notify guardians or authorities when the SOS button is activated. This method ensures that help can be requested quickly with minimal interaction from the user.

Several studies have also explored the use of smartphone sensors and trigger mechanisms such as shake detection, power button presses, or voice commands to activate emergency alerts. These features allow users to send alerts discreetly without unlocking the phone or opening the application, which can be extremely useful in dangerous situations.

In addition to alert systems, many safety applications incorporate multimedia evidence collection features such as camera and audio recording. Capturing photos or audio during emergencies can provide useful evidence and help authorities understand the situation more clearly. Modern smartphones make it possible to quickly capture and share such information through messaging or social media platforms.

From a technological perspective, mobile applications commonly use Android development frameworks, programming languages such as Java or Kotlin, and cloud-based services such as Firebase for authentication and database management. Firebase provides secure user authentication, real-time data storage, and easy integration with mobile applications, making it a popular choice for mobile application development.

The combination of GPS tracking, SOS alerts, multimedia capture, and cloud-based data management forms the foundation of many existing women safety applications.

2.2 Gap Analysis

Feature	Traditional Safety Apps	Shyra Mobile App	Gap Resolved
SOS Alert System	Provides basic SOS button that sends a simple message to contacts.	Provides a one-tap SOS button that sends an SMS along with a Google Maps live location link.	Improves emergency communication by providing accurate location details to contacts.
Location Sharing	Some apps provide location sharing but often depend on internet connectivity.	Uses GPS to obtain live location and sends it through SMS with a map link.	Allows location sharing even with limited internet connectivity.
Emergency Contact Management	Limited contact management or fixed contacts.	Users can add, edit, and delete multiple emergency contacts easily.	Provides flexible and user-controlled contact management.
Discreet Emergency Trigger	Most apps require the user to open the app and press SOS.	SOS can be triggered by pressing the power button multiple times.	Allows users to trigger emergency alerts even when the phone is locked or when the app cannot be opened.
Evidence Collection	Many apps do not support capturing evidence during emergencies.	Supports camera photo capture and audio recording features.	Enables users to collect evidence that may help in investigations.
User Interface	Some safety apps have complex interfaces that are difficult to use during emergencies.	Simple and user-friendly interface with clearly labelled safety features.	Makes the application easier and faster to use in stressful situations.
Privacy Protection	Some applications store media or personal data on cloud servers.	Media files are stored locally on the device and shared only when the user chooses.	Improves user privacy and prevents unauthorized access to personal media.
Additional Safety Information	Many apps focus only on alerts and do not provide safety guidance.	Includes a Safety Tips section to educate users about personal safety practices.	Increases user awareness and preparedness for unsafe situations.

1.1 Gap analysis between traditional system and our proposed system

The gap analysis clearly shows that the Shyra Women Safety Application addresses several limitations found in traditional safety applications by integrating multiple emergency features, improving usability, and enhancing privacy protection.

2.3 Summary of the Findings

The literature survey shows that mobile technology can play a significant role in improving personal safety through features such as location tracking, emergency alerts, and multimedia communication. Existing research and applications demonstrate that integrating smartphone capabilities with communication systems can help users quickly notify others and receive assistance during emergencies.

However, many current applications lack a comprehensive set of features that combine emergency alerts, location sharing, evidence collection, and discreet activation methods in a single platform. Additionally, issues related to usability, network dependency, and privacy protection still need improvement.

Based on these observations, the Shyra Women Safety Application is designed to address these gaps by integrating multiple safety features into a single Android application. The system provides SOS alerts with live location sharing, emergency contact management, camera and audio evidence capture, and a power-button based emergency trigger. The application also focuses on providing a simple and user-friendly interface while ensuring user privacy by storing media files locally.

Therefore, the proposed system aims to provide a more complete and practical safety solution that can assist users effectively during emergency situations.

CHAPTER-3

SYSTEM REQUIREMENTS

SYSTEM REQUIREMENTS

3.1 System Requirements

System requirements describe the hardware and software components needed to develop and run the Shyra Women Safety Application. These requirements ensure that the application functions properly and provides reliable performance on supported devices.

The system requirements are divided into two categories: hardware requirements and software requirements.

3.1.1 Hardware Requirements

The hardware requirements specify the physical devices needed for developing and using the application. The following hardware components are required for the development and operation of the system:

Hardware Component	Requirement
Processor	Intel Core i3 or higher
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 10 GB free disk space
Mobile Device	Android smartphone
GPS Sensor	Required for location tracking
Camera	Required for capturing photos
Microphone	Required for audio recording
Internet Connection	Required for Firebase authentication and database access

1.2 Hardware Requirements

The development of the application is performed on a computer or laptop with sufficient processing power and memory to run Android development tools. An Android smartphone is required to test the application features such as GPS location tracking, camera capture, and SOS alerts.

3.1.2 Software Requirements

Software requirements refer to the programs and tools needed to develop, run, and manage the application. The Shyra Women Safety Application is built using modern Android development tools and services.

Software Component	Description
Operating System	Windows 10 / Windows 11
Programming Language	Java
Development Environment	Android Studio
UI Design	XML Layouts
Backend Services	Firebase Authentication, Firebase Realtime Database
Android SDK	Android SDK API Level 21 or above
Location Services	Google Play Services (Fused Location Provider)
Database	Firebase Realtime Database
Version Control (Optional)	Git / GitHub

1.3 Software Requirements

Android Studio is used as the main development environment for designing the user interface, writing the application code, and testing the application. Firebase services are used to manage user authentication and store emergency contact data securely. The Android SDK provides the required libraries and tools for building Android applications.

These hardware and software requirements ensure that the system can be developed efficiently and run smoothly on supported Android devices.

3.2 Android Studio



Fig 1.1 Android Studio

Android Studio is the official Integrated Development Environment (IDE) for Android application development, provided by Google. It is widely used for designing, developing, testing, and debugging Android applications. In the development of the Shyra Women Safety Application, Android Studio played a crucial role in implementing all features and managing the overall project structure.

Android Studio provides a powerful and flexible environment that supports the complete Android development lifecycle. It includes a rich code editor, intelligent code completion, real-time error detection, and built-in tools for debugging and performance analysis. The IDE also offers seamless integration with the Android SDK, making it easier to develop applications that are compatible with various Android devices.

One of the key advantages of Android Studio is its layout editor, which allows developers to design user interfaces using XML. The drag-and-drop feature helps in creating visually appealing layouts, while the code editor allows precise customization of UI components. In this project, XML layouts were used to design screens such as login, dashboard, emergency contacts, safety tips, and settings.

Another important feature is the Android Emulator, which allows developers to test applications without requiring a physical device. However, in this project, both emulator and real Android devices were used to ensure accurate testing of features such as GPS location, SMS functionality, camera, and audio recording.

The IDE includes a Logcat tool that helps in debugging by displaying system logs and error messages in real time. This tool was extensively used to identify and fix issues related to permissions, location services, and Firebase integration.

Android Studio also supports direct integration with Firebase services, making it easier to implement features such as user authentication and real-time database storage. In this project, Firebase was connected to Android Studio to manage user accounts and store emergency contact data securely.

3.3 Firebase in the Project

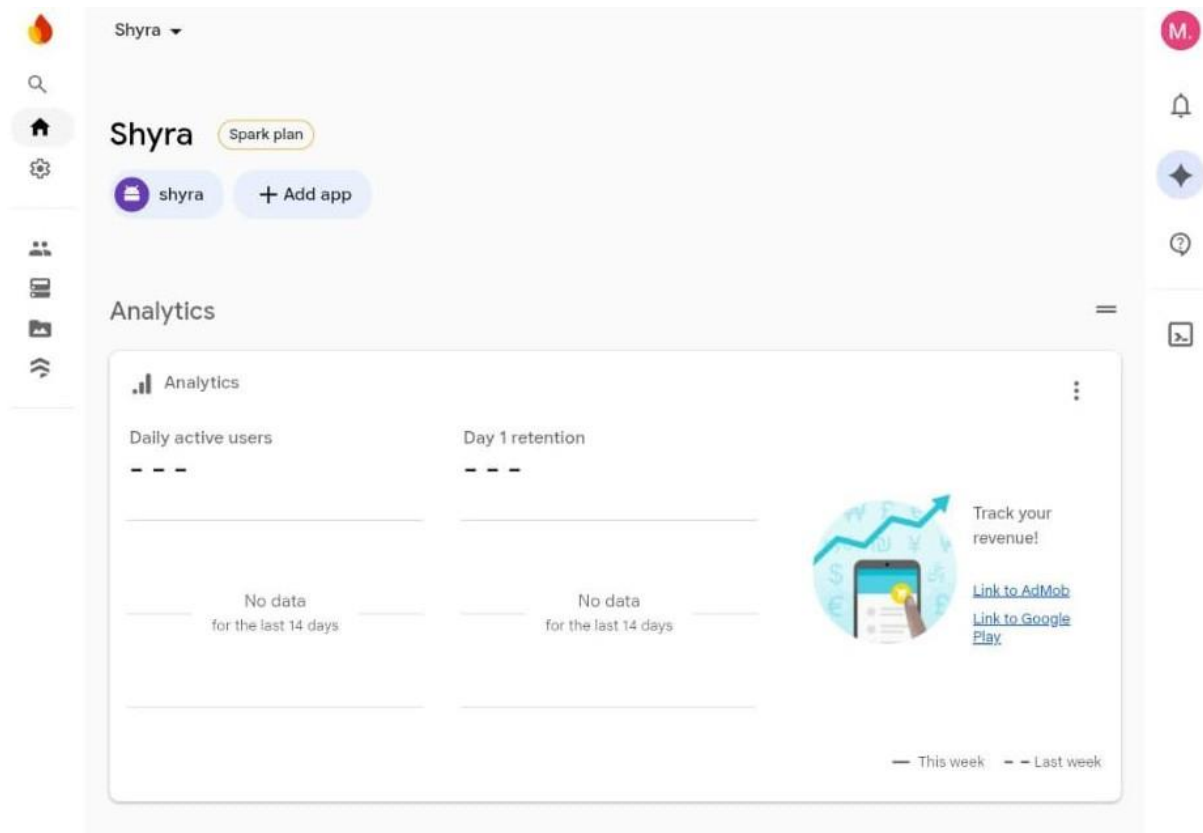


Fig 1.2 Firebase Console

Firebase is a Backend-as-a-Service (BaaS) platform provided by Google that offers various tools for building scalable and secure mobile applications. In the Shyra Women Safety Application, Firebase is used to handle user authentication and real-time data storage efficiently.

The application uses Firebase Authentication to manage user registration and login processes. Users can create an account using their email and password, and Firebase ensures secure authentication by handling credentials safely. This eliminates the need to build a custom authentication system from scratch.

The application also uses the Firebase Realtime Database to store and manage user data such as emergency contacts. Each user's data is stored in a structured format, allowing quick retrieval and updates. Since it is a real-time database, any changes made to the data are instantly reflected without requiring manual refresh.

Firebase integration in Android Studio simplifies backend development by providing ready-to-use APIs. It ensures data security, scalability, and fast performance. By using Firebase, the application achieves reliable data management and seamless synchronization between the app and the database.

3.4 Fused Location Provider

The Fused Location Provider API, part of Google Play Services, is used in the application to obtain accurate and efficient location data. It combines multiple location sources such as GPS, Wi-Fi, and mobile networks to provide the best possible location information.

In the Shyra application, the Fused Location Provider is used to fetch the user's current location when an SOS alert is triggered. The system retrieves latitude and longitude coordinates and converts them into a Google Maps link, which is then sent to emergency contacts via SMS.

The key advantage of using the Fused Location Provider is its ability to balance accuracy and battery efficiency. Unlike traditional GPS-only methods, it intelligently selects the best source of location data, reducing battery consumption while maintaining precision.

The API also allows developers to set parameters such as location update intervals, priority (high accuracy or low power), and distance thresholds. This ensures that the application provides timely and accurate location updates during emergency situations.

3.5 Accessibility Service (Power Button SOS)

The Accessibility Service in Android is designed to assist users with disabilities by allowing applications to monitor and respond to system-level events. In the Shyra Women Safety Application, this feature is utilized to implement the power button SOS trigger.

The application detects multiple presses of the device's power button within a short time interval. When this pattern is recognized, the app automatically triggers an SOS alert without requiring the user to unlock the phone or open the application. This is particularly useful in dangerous situations where quick and discreet action is required.

The Accessibility Service runs in the background and listens for specific hardware interactions. Once the SOS trigger condition is met, the application performs actions such as sending emergency messages, sharing location, and initiating recording features.

However, due to security and privacy restrictions in Android, users must manually enable the Accessibility Service for the application in device settings. Additionally, the behavior of this feature may vary across different devices due to manufacturer-specific customizations.

Despite these limitations, the Accessibility Service provides an effective and innovative way to trigger emergency alerts quickly, making it a crucial feature of the Shyra Women Safety Application.

CHAPTER 4

SYSTEM ANALYSIS AND DESIGN

SYSTEM ANALYSIS AND DESIGN

4.1 Requirement Analysis

Requirement analysis is the process of identifying and defining the needs and expectations of users for the proposed system. For the Shyra Women Safety Application, the primary requirement is to provide a reliable and easy-to-use mobile platform that helps women quickly seek assistance during emergency situations.

The system must allow users to register and log in securely, store emergency contact information, and send SOS alerts with their live location to trusted contacts. The application should also allow users to capture photos and record audio that may serve as evidence during unsafe situations. Additionally, the system must provide location-sharing features and safety tips to improve user awareness.

The application should work smoothly on Android smartphones and must provide quick response during emergencies. The user interface should be simple and easy to understand so that users can operate the application quickly even in stressful situations.

4.2 Functional Requirements

Functional requirements describe the main features and operations that the system must perform.

1. The system must allow users to register and log in using email and password.
2. The system must allow users to add, edit, and delete emergency contacts.
3. The system must provide an SOS button that sends emergency alerts to saved contacts.
4. The system must send the user's live GPS location along with the SOS message.
5. The system must allow users to share their current location through messaging applications.
6. The system must allow users to capture photos using the device camera.
7. The system must allow users to record audio using the device microphone.
8. The system must allow users to share captured media through installed apps.
9. The system must support a power-button based SOS trigger for emergency activation.
10. The system must display safety tips and awareness information.
11. The system must allow users to log out and manage settings.

4.3 Non-Functional Requirements

Non-functional requirements describe the performance and quality attributes of the system.

1. Usability

The application must provide a simple and user-friendly interface so that users can easily navigate and use the features during emergencies.

2. Reliability

The system should function consistently without crashes and should send alerts reliably during emergencies.

3. Performance

The application must respond quickly when the SOS button is activated and retrieve location data efficiently.

4. Security

User authentication must be secure. Access to personal data such as contacts should be restricted to authorized users only.

5. Privacy

Sensitive media files such as photos and audio recordings should be stored locally to protect user privacy.

6. Compatibility

The system should work on Android devices running API level 21 or above.

4.4 System Architecture

The system follows a client-server architecture. The Android mobile application acts as the client, while Firebase services act as the backend server.

The architecture consists of three main layers:

1. Presentation Layer

This layer contains the user interface designed using XML layouts. It includes screens such as login, home dashboard, emergency contacts, safety tips, and settings.

2. Application Logic Layer

This layer contains the core functionality implemented using Java. It handles user authentication, SOS alerts, location retrieval, media capture, and interaction with Firebase services.

3. Data Layer

This layer includes Firebase Authentication for user login and Firebase Realtime Database for storing emergency contact information. Media files are stored locally on the device.

4.5 Entity-Relationship (ER) Diagram

The ER diagram represents the relationship between different entities in the system.

Main entities in the system include:

User

- UserID (Primary Key)
- Name
- Email
- Password

EmergencyContact

- ContactID (Primary Key)
- Name
- PhoneNumber
- Relationship
- UserID (Foreign Key)

A User can have multiple Emergency Contacts, creating a one-to-many relationship between User and EmergencyContact entities.

The user interacts with the system to perform these actions, and the system responds accordingly.

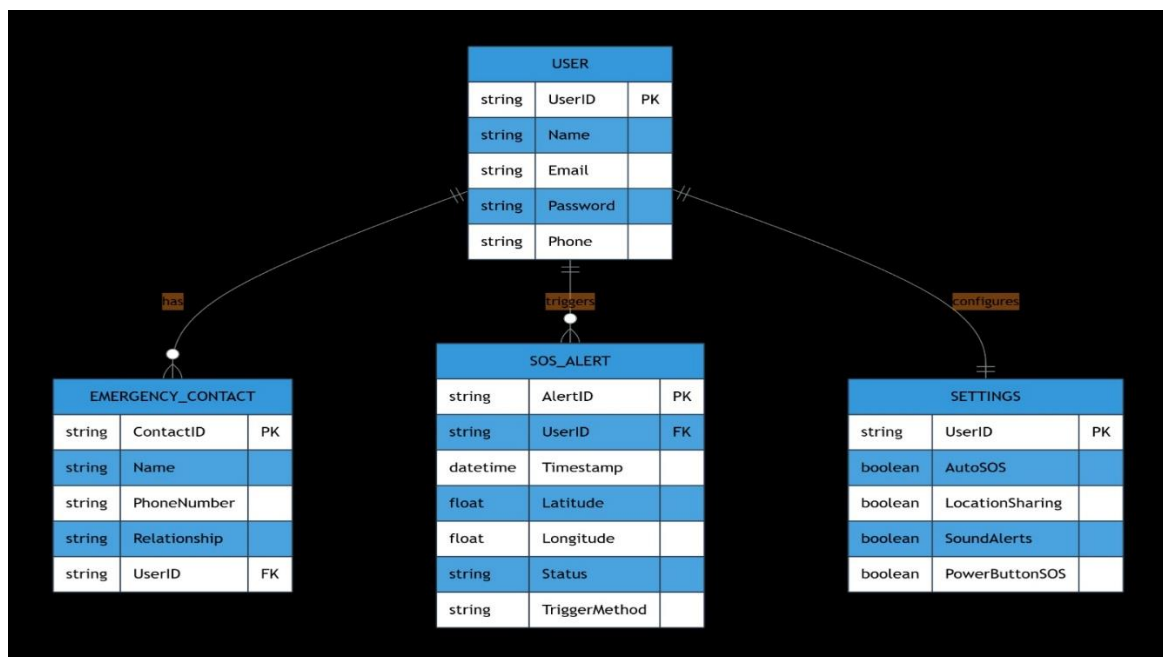


Fig 1.3 E-R Diagram

4.6 Use Case Diagram

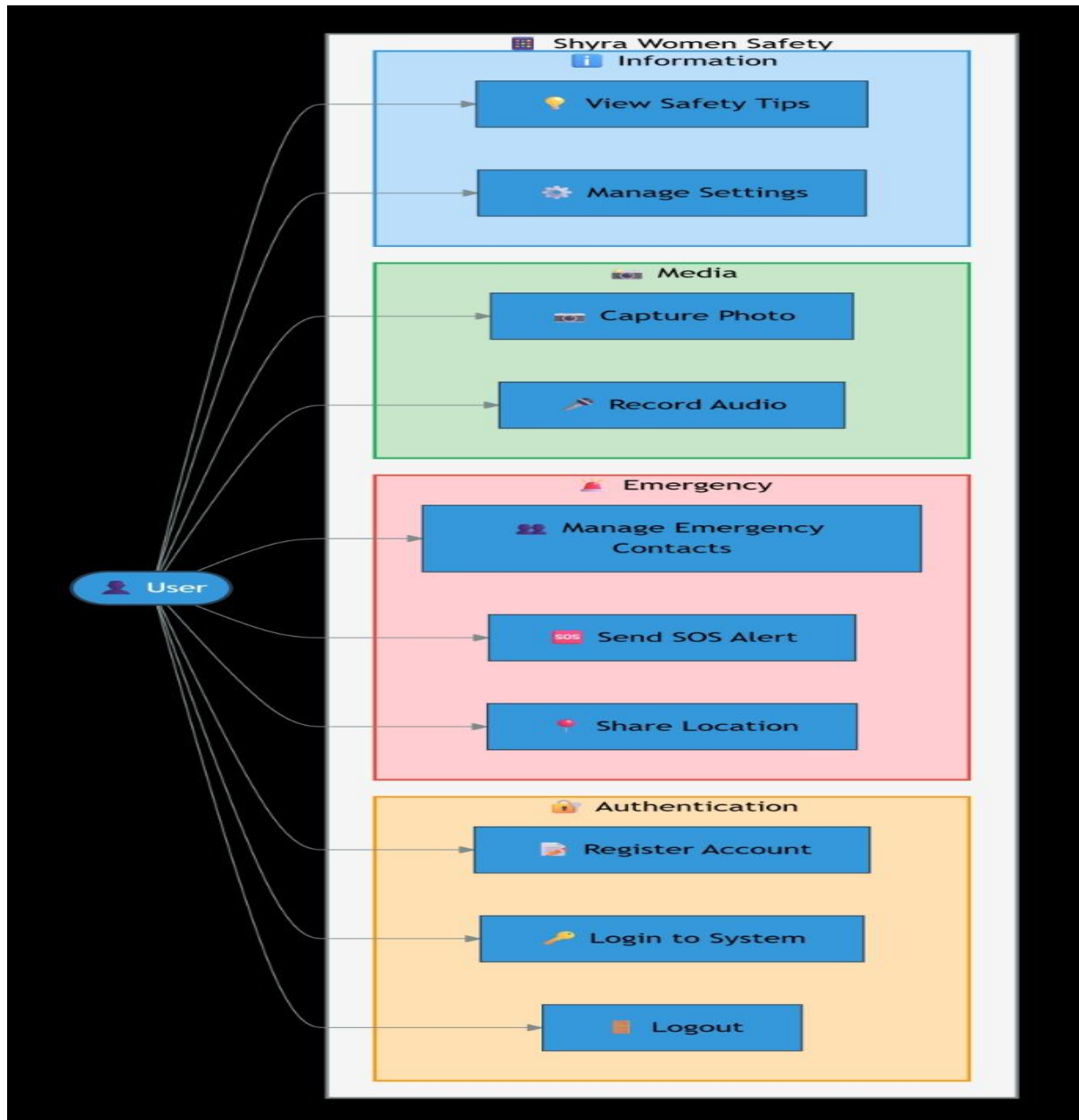


Fig 1.4 Use Case Diagram

4.7 Data Flow Diagram – Level 0

The Level 0 Data Flow Diagram represents the overall flow of information in the system.

Input:

User provides login credentials, emergency contact details, and SOS trigger actions.

Process:

The system authenticates the user, retrieves location data, processes emergency alerts, and manages contacts.

Output:

SOS alerts with location information are sent to emergency contacts, and media files are captured or shared when required.

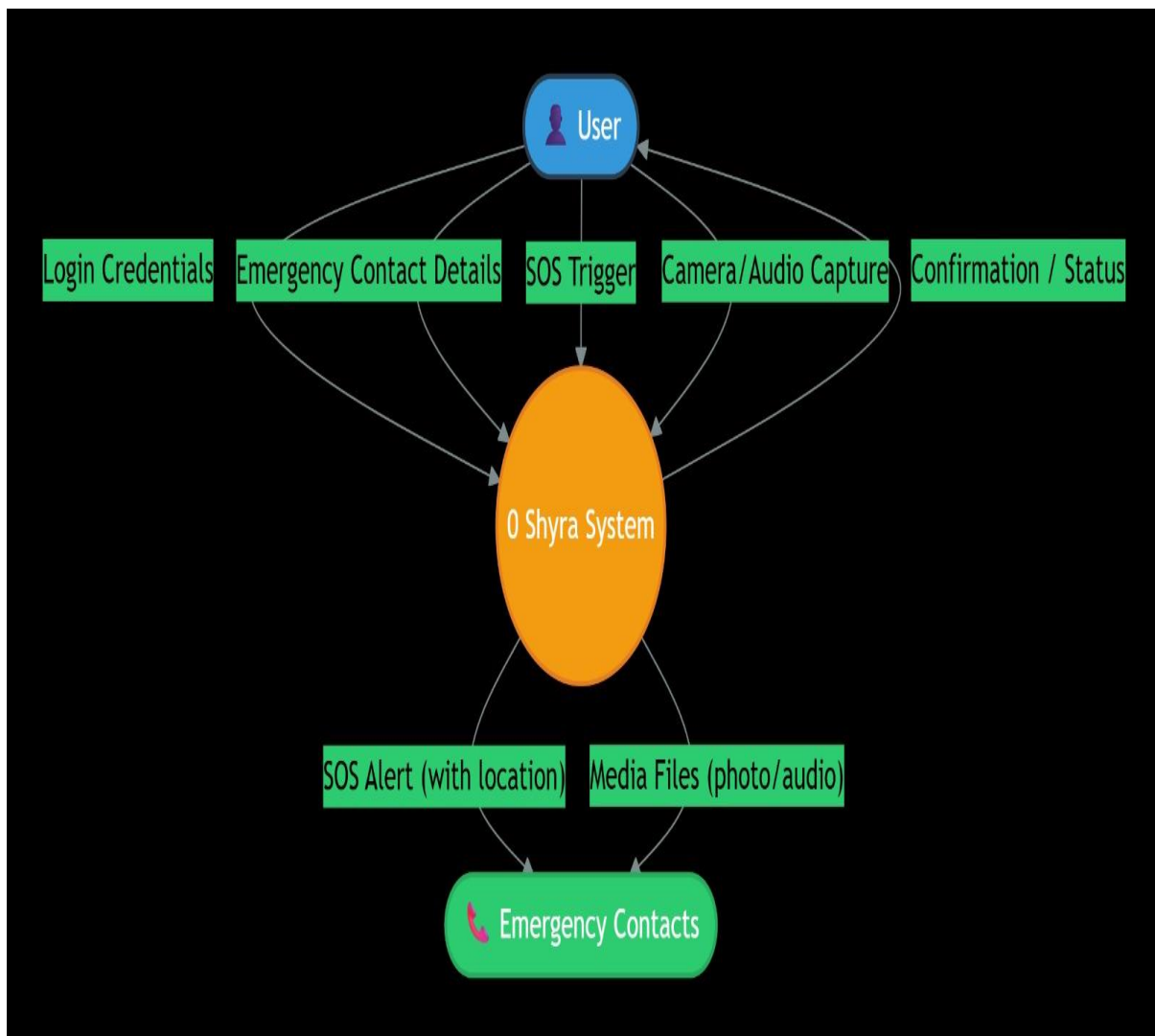


Fig 1.5 Data Flow Diagram – Level 0

4.8 Data Flow Diagram – Level 1

The Level 1 Data Flow Diagram provides a more detailed representation of system processes.

Main processes include:

1. User Authentication Process
 - User enters email and password
 - Firebase verifies credentials
 - User gains access to the application
2. Contact Management Process
 - User adds or edits emergency contacts
 - Contact information is stored in Firebase database
3. SOS Alert Process
 - User presses SOS button or power button trigger
 - System retrieves GPS location
 - SMS message with location link is sent to emergency contacts
4. Media Capture Process
 - User captures photo or records audio
 - Media is stored locally and shared via messaging apps

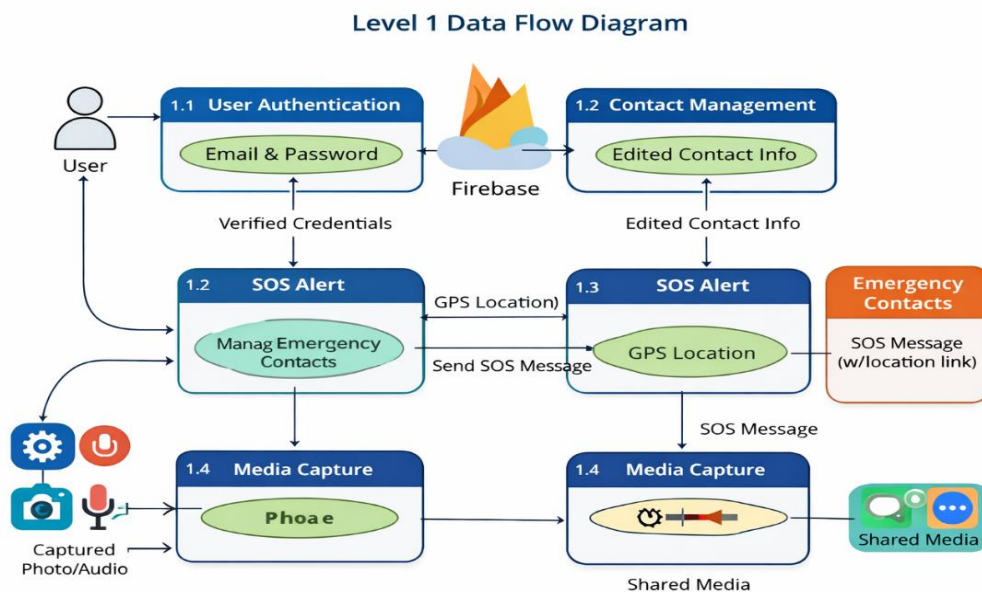


Fig 1.6 Data Flow Diagram – Level 1

4.9 Flow Chart

The system flow begins when the user launches the application.

1. User opens the application.
2. If the user is not logged in, the system displays the login or registration screen.
3. After successful login, the user is taken to the main dashboard.
4. The user can perform actions such as managing contacts, sending SOS alerts, sharing location, or capturing media.
5. If SOS is triggered, the system retrieves the user's location and sends alerts to emergency contacts.
6. The user can log out or close the application.

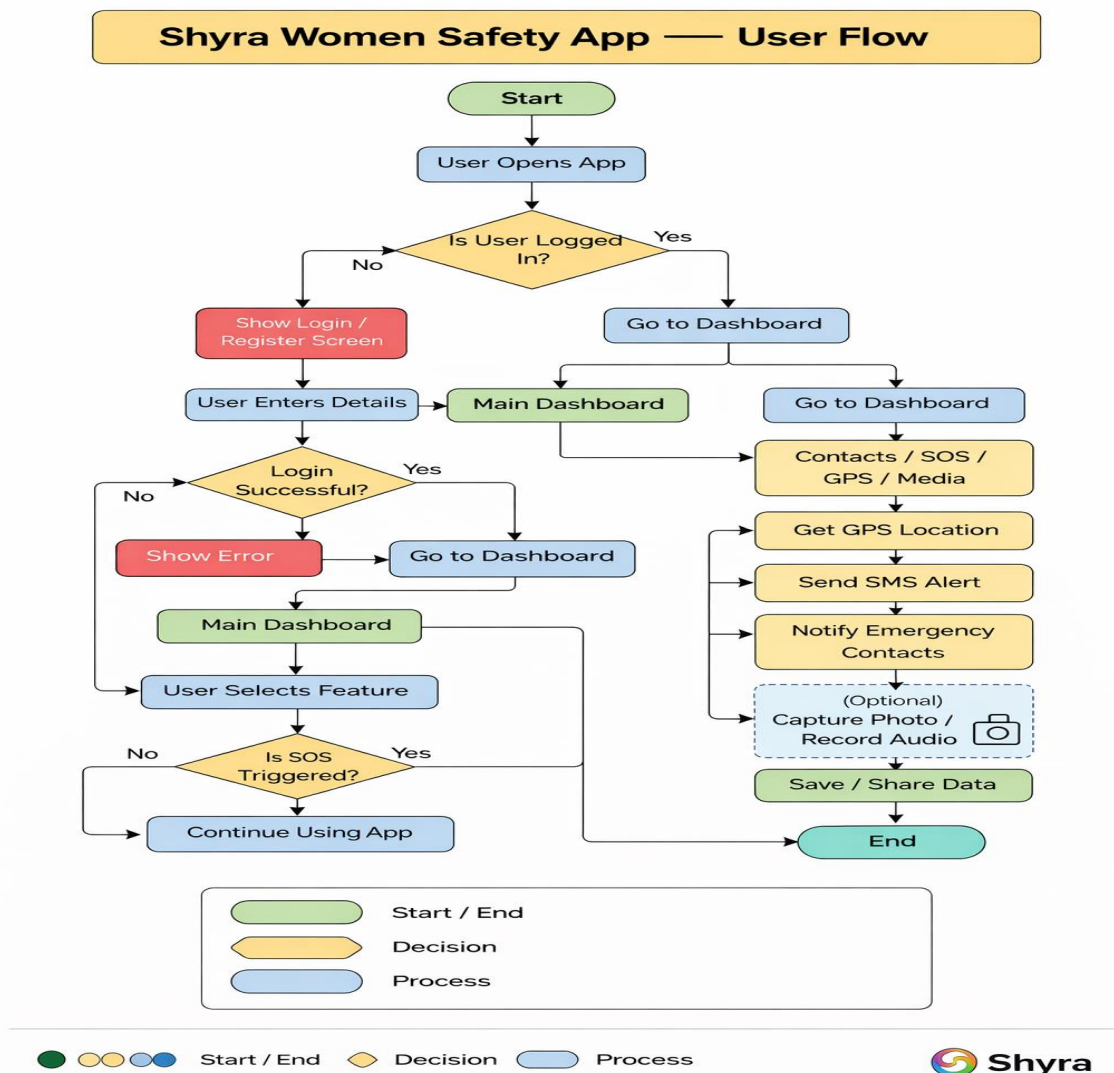


Fig 1.7 Flow Chart

CHAPTER-5

IMPLEMENTATION

IMPLEMENTATION

5.1 Objectives of the Implementation

The implementation phase focuses on converting the system design into a fully functional application. The main objective of implementing the Shyra Women Safety Application is to develop a reliable Android application that provides quick emergency assistance and safety features for users.

The implementation ensures that all designed modules such as user authentication, emergency contact management, SOS alerts, GPS location sharing, and media capture are properly developed and integrated. The system is implemented in a way that ensures quick response during emergencies, smooth navigation, and secure handling of user information.

Another objective is to create a user-friendly interface that allows users to access important safety features easily without requiring technical knowledge. The implementation also focuses on ensuring system stability, data security, and compatibility with different Android devices.

5.2 Module-Based Implementation Approach

The application is implemented using a modular development approach, where each major functionality is developed as a separate module. This approach improves maintainability, scalability, and ease of debugging.

The main modules implemented in the system include:

1. **User Authentication Module:** Handles user registration, login, and session management using Firebase Authentication.
2. **Emergency Contact Module:** Allows users to add, edit, and delete emergency contacts stored in Firebase Realtime Database.
3. **SOS Alert Module:** Sends emergency SMS alerts with the user's live GPS location to all saved emergency contacts.
4. **Location Sharing Module:** Retrieves the user's current location using the GPS system and allows sharing through messaging applications.
5. **Camera Module:** Captures photos using the device camera and allows instant sharing.
6. **Audio Recording Module:** Records audio using the MediaRecorder API and allows sharing or local storage.
7. **Power Button SOS Module:** Detects multiple power button presses using an Accessibility Service and triggers emergency alerts.
8. **Safety Tips Module:** Displays safety awareness tips to educate users about personal safety practices.

5.3 Data Flow during Implementation

The data flow in the application begins when the user interacts with the system. The user first registers or logs in using their credentials. These credentials are verified through Firebase Authentication.

Once logged in, the user can manage emergency contacts, which are stored in the Firebase Realtime Database under the user's unique ID. When the SOS button is pressed, the application retrieves the user's location through the GPS service and generates a Google Maps link.

The application then sends an SMS containing the location link to all stored emergency contacts. If the user captures photos or records audio, the media files are stored locally on the device and can be shared through messaging applications.

5.4 Implementation Strategy

The development of the Shyra application followed a step-by-step implementation strategy:

1. Setting up the Android development environment.
2. Designing the user interface layouts using XML.
3. Implementing Firebase authentication for user login and registration.
4. Creating the emergency contact management system.
5. Integrating GPS location services.
6. Implementing the SOS alert messaging system.
7. Developing camera and audio recording modules.
8. Implementing power button SOS functionality.
9. Testing and debugging the application.
10. Final integration and performance optimization.

5.5 Features Implemented

The final application includes several important safety features:

- User registration and login
- Emergency contact management
- SOS emergency alert system
- Live GPS location sharing
- Camera photo capture
- Audio recording and sharing

- Power button triggered SOS alerts
- Safety tips section
- Navigation drawer and swipe navigation
- Settings and logout functionality

These features work together to provide a complete women safety solution.

5.6 Frontend vs Backend Functional Comparison

Frontend (User Interface)	Backend (System Processing)
Login and registration screens	Firebase Authentication verification
Contact management interface	Firebase database storage
SOS button and navigation interface	SMS alert processing
Camera and audio capture UI	Media handling and file storage
Safety tips and settings screens	Location retrieval and system services

1.4 Frontend vs Backend Functional Comparison

The frontend handles user interaction, while the backend manages authentication, data storage, and system services.

5.7 Final System Structure

The final system structure consists of multiple components working together to provide the complete application.

- User Interface Layer – Displays screens and handles user interaction.
- Application Logic Layer – Processes user actions and implements application features.
- Firebase Backend Layer – Manages authentication and database operations.
- Device Services Layer – Handles GPS, camera, microphone, and SMS services.

These layers work together to ensure smooth functionality of the application.

5.8 Summary of Implementation

The implementation of the Shyra Women Safety Application successfully transforms the system design into a working Android application. All the modules were implemented and integrated effectively to ensure smooth performance and reliability.

The application provides essential safety features such as emergency alerts, live location sharing, and media capture. The modular design allows the system to be easily maintained and upgraded with additional features in the future.

5.9 Detailed Implementation Process

The detailed implementation process of the Shyra Women Safety Application involved multiple stages of development, testing, and refinement. The application was developed using a structured approach to ensure that each feature was implemented correctly and integrated smoothly with other modules.

The first stage of implementation involved setting up the development environment. Android Studio was installed as the primary development tool, and the Android SDK was configured to support devices running Android API level 21 and above. Firebase services were also integrated into the project by connecting the Android application to a Firebase project through the Firebase console. This enabled the use of Firebase Authentication and Firebase Realtime Database within the application.

The next stage involved designing the user interface (UI) using XML layout files. Multiple screens were created for different functionalities such as login, registration, main dashboard, emergency contact management, safety tips, and settings. Material Design principles were followed to ensure a modern and user-friendly interface. Navigation components such as the Navigation Drawer and ViewPager2 were implemented to allow smooth navigation between different sections of the application.

After completing the UI design, the user authentication module was implemented. Firebase Authentication was integrated to allow users to register and log in using their email and password. The system verifies the user credentials through Firebase servers and maintains a user session so that the user does not need to log in repeatedly.

The next module implemented was the Emergency Contact Management System. This module allows users to add, edit, and delete emergency contacts. The contact details such as name, phone number, and relationship are stored in Firebase Realtime Database under the user's unique ID. This ensures that each user can access only their own contact information.

Following this, the SOS alert module was developed. The SOS feature includes a prominent emergency button on the main dashboard. When the user presses this button, the system retrieves the user's current location using the Google Play Services Fused Location Provider API. The location coordinates are then converted into a Google Maps link, which is included in an SMS message sent to all stored emergency contacts. This feature ensures that trusted individuals can quickly locate the user during emergencies.

The location sharing module was also implemented to allow users to manually retrieve and share their current GPS location with others through messaging applications such as SMS or WhatsApp. The application also checks whether GPS services are enabled and prompts the user to enable location services if necessary.

Another important part of the implementation was the camera and audio recording module. The camera feature allows users to capture photos using the device's default camera application. The captured images can then be shared immediately through installed applications. Similarly, the audio recording feature uses the MediaRecorder API to record audio files in AAC format. These recordings can be shared with contacts or saved locally on the device if sharing is not possible.

The Power Button SOS module was implemented using an Android Accessibility Service. This service monitors power button presses and detects rapid multiple presses. When the power button is pressed several times in quick succession, the system triggers an SOS alert even if the phone screen is locked. This feature provides a discreet method of activating emergency alerts.

Finally, permission handling was implemented using the Dexter library to manage runtime permissions such as location access, SMS sending, camera usage, audio recording, and contacts access. Proper permission management ensures that the application works correctly while complying with Android security policies.

After implementing all modules, the application underwent extensive testing and debugging to ensure that all features worked properly under different conditions.

5.10 System Integration and Refinement

System integration is the process of combining all individual modules of the application into a single functioning system. In the Shyra Women Safety Application, several modules such as authentication, contact management, location tracking, SOS alerts, and media capture were developed independently and then integrated together to form the final application.

The integration process began by connecting the authentication module with the main application dashboard. After successful login, the user is redirected to the main dashboard where all safety features are accessible. The authentication system also ensures that user-specific data, such as emergency contacts, are retrieved correctly from the Firebase database.

Next, the contact management module was integrated with the SOS alert system. When an SOS alert is triggered, the system retrieves all emergency contacts stored in the database and sends alert messages to them. This integration ensures that emergency alerts are delivered to the correct recipients.

The location services module was integrated with the SOS module and the location sharing feature. The application retrieves accurate GPS coordinates and converts them into a Google Maps link before sending them through SMS or other messaging platforms. This integration ensures that location information is always included in emergency alerts.

The camera and audio modules were integrated with the Android sharing system using intents. This allows users to quickly share captured photos or audio recordings through installed

applications such as WhatsApp, Telegram, or email. A fallback mechanism was also implemented to store the files locally in the Downloads

5.11 Code

The implementation of the Shyra application involves several Java classes and XML layout files. The main code components include:

- LoginActivity.java – Handles user authentication.
- MainActivity.java – Displays the main dashboard.
- ContactActivity.java – Manages emergency contacts.
- PowerButtonSOSService.java – Detects power button presses.
- LocationHelper.java – Retrieves GPS location.
- FirebaseHelper.java – Handles Firebase operations.

These code modules interact with Android system services and Firebase backend services to provide the complete functionality of the women safety application.

LoginActivity.java

```
package com.example.shyra.activities;

import android.content.Intent;
import android.os.Bundle;
import android.text.TextUtils;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ProgressBar;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import com.example.shyra.R;
import com.google.firebase.auth.FirebaseAuth;

public class LoginActivity extends AppCompatActivity {

    private EditText emailEditText, passwordEditText;
    private Button loginButton;
    private TextView registerLink;
    private ProgressBar progressBar;
```

```

private FirebaseAuth mAuth;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_login);

    mAuth = FirebaseAuth.getInstance();

    initViews();
    setupListeners();
}

private void initViews() {
    emailEditText = findViewById(R.id.editTextEmail);
    passwordEditText = findViewById(R.id.editTextPassword);
    loginButton = findViewById(R.id.buttonLogin);
    registerLink = findViewById(R.id.textViewRegister);
    progressBar = findViewById(R.id.progressBar);
}

private void setupListeners() {
    loginButton.setOnClickListener(v -> attemptLogin());
    registerLink.setOnClickListener(v -> {
        startActivity(new Intent(LoginActivity.this, RegisterActivity.class));
    });
}

private void attemptLogin() {
    String email = emailEditText.getText().toString().trim();
    String password = passwordEditText.getText().toString().trim();

    if (TextUtils.isEmpty(email)) {
        emailEditText.setError("Email is required");
        emailEditText.requestFocus();
        return;
    }

    if (TextUtils.isEmpty(password)) {
        passwordEditText.setError("Password is required");
        passwordEditText.requestFocus();
        return;
    }

    progressBar.setVisibility(View.VISIBLE);
    loginButton.setEnabled(false);
}

```

```

        mAuth.signInWithEmailAndPassword(email, password)
            .addOnCompleteListener(this, task -> {
                progressBar.setVisibility(View.GONE);
                loginButton.setEnabled(true);

                if (task.isSuccessful()) {
                    Toast.makeText(LoginActivity.this, "Login successful",
                        Toast.LENGTH_SHORT).show();
                    startActivity(new Intent(LoginActivity.this, MainActivity.class));
                    finish();
                } else {
                    String errorMessage = task.getException() != null ?
                        task.getException().getMessage() : "Authentication failed";
                    Toast.makeText(LoginActivity.this, "Login failed: " + errorMessage,
                        Toast.LENGTH_LONG).show();
                }
            });
    }
}

```

MainActivity.java

```

package com.example.shyra.activities;

import android.content.Intent;
import android.os.Bundle;
import android.view.MenuItem;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.ActionBarDrawerToggle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import androidx.core.view.GravityCompat;
import androidx.drawerlayout.widget.DrawerLayout;
import androidx.viewpager2.widget.ViewPager2;

import com.example.shyra.R;
import com.example.shyra.adapters.MainPagerAdapter;
import com.google.android.material.floatingactionbutton.FloatingActionButton;
import com.google.android.material.navigation.NavigationView;
import com.google.firebase.auth.FirebaseAuth;

```

```

public class MainActivity extends AppCompatActivity implements
NavigationView.OnNavigationItemSelectedListener {

    private DrawerLayout drawerLayout;
    private NavigationView navigationView;
    private Toolbar toolbar;
    private ViewPager2 viewPager;
    private FloatingActionButton profileButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        toolbar = findViewById(R.id.toolbar);
        drawerLayout = findViewById(R.id.drawer_layout);
        navigationView = findViewById(R.id.nav_view);
        viewPager = findViewById(R.id.viewPager);
        profileButton = findViewById(R.id.buttonProfile);

        setSupportActionBar(toolbar);
        ActionBarDrawerToggle toggle = new ActionBarDrawerToggle(
            this, drawerLayout, toolbar, R.string.navigation_drawer_open,
            R.string.navigation_drawer_close);
        drawerLayout.addDrawerListener(toggle);
        toggle.syncState();
        navigationView.setNavigationItemSelectedListener(this);

        // Set up ViewPager
        viewPager.setAdapter(new MainPageAdapter(this));

        // Profile button click
        profileButton.setOnClickListener(v -> openProfile());
    }

    @Override
    public boolean onNavigationItemSelectedListener(@NonNull MenuItem item) {
        int id = item.getItemId();
        if (id == R.id.nav_profile) {
            openProfile();
        } else if (id == R.id.nav_settings) {
            startActivity(new Intent(this, SettingsActivity.class));
        } else if (id == R.id.nav_logout) {
            FirebaseAuth.getInstance().signOut();
            Toast.makeText(this, "Logged out", Toast.LENGTH_SHORT).show();
            startActivity(new Intent(this, LoginActivity.class));
        }
    }
}

```

```

        finish();
    } else if (id == R.id.nav_safety_tips) {
        viewPager.setCurrentItem(1);
    } else if (id == R.id.nav_about) {
        viewPager.setCurrentItem(2);
    }
    drawerLayout.closeDrawer(GravityCompat.START);
    return true;
}

private void openProfile() {
    startActivity(new Intent(this, ProfileActivity.class));
}
}

```

PowerButtonSOSService.java

```

package com.example.shyra.services;

import android.accessibilityservice.AccessibilityService;
import android.accessibilityservice.GestureDescription;
import android.app.Notification;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Intent;
import android.media.MediaPlayer;
import android.media.RingtoneManager;
import android.net.Uri;
import android.os.Build;
import android.os.Handler;
import android.os.Looper;
import android.os.PowerManager;
import android.os.Vibrator;
import android.util.Log;
import android.view.KeyEvent;
import android.view.accessibility.AccessibilityEvent;

import androidx.core.app.NotificationCompat;

import com.example.shyra.R;
import com.example.shyra.activities.MainActivity;

public class PowerButtonSOSService extends AccessibilityService {

```

```

private static final String TAG = "PowerButtonSOS";
private static final int PRESS_COUNT_THRESHOLD = 5;
private static final long PRESS_WINDOW_MS = 3000; // 3 seconds
private static final long SOS_DELAY_MS = 5000; // 5-second countdown before sending

private int pressCount = 0;
private final Handler handler = new Handler(Looper.getMainLooper());
private final Runnable resetPressCountRunnable = this::resetPressCount;

private boolean isCountdownActive = false;
private Runnable sosRunnable;
private MediaPlayer alarmPlayer;
private Vibrator vibrator;

@Override
public void onCreate() {
    super.onCreate();
    vibrator = (Vibrator) getSystemService(VIBRATOR_SERVICE);
    createNotificationChannel();
    startForegroundServiceWithNotification();
}

@Override
public void onAccessibilityEvent(AccessibilityEvent event) {
    // Not used for key events, but required by AccessibilityService
}

@Override
protected boolean onKeyEvent(KeyEvent event) {
    if (event.getKeyCode() == KeyEvent.KEYCODE_POWER) {
        if (event.getAction() == KeyEvent.ACTION_DOWN) {
            pressCount++;
            Log.d(TAG, "Power press count: " + pressCount);

            handler.removeCallbacks(resetPressCountRunnable);
            handler.postDelayed(resetPressCountRunnable, PRESS_WINDOW_MS);

            if (pressCount >= PRESS_COUNT_THRESHOLD && !isCountdownActive) {
                pressCount = 0;
                handler.removeCallbacks(resetPressCountRunnable);
                startSOSCountdown();
            }
        }
        return true;
    }
}

```



```

        return super.onKeyEvent(event);
    }

    private void resetPressCount() {
        pressCount = 0;
        Log.d(TAG, "Press count reset");
    }

    private void startSOSCountdown() {
        isCountdownActive = true;

        playAlarm();
        vibratePattern();
        wakeUpScreen();

        showSOSNotification("SOS will be sent in 5 seconds. Tap to cancel.");

        sosRunnable = () -> {
            stopAlarm();
            isCountdownActive = false;
            sendSOS();
        };
        handler.postDelayed(sosRunnable, SOS_DELAY_MS);
    }

    private void cancelSOSCountdown() {
        if (isCountdownActive) {
            handler.removeCallbacks(sosRunnable);
            stopAlarm();
            isCountdownActive = false;
            showNotification("SOS cancelled", true);
            Log.d(TAG, "SOS cancelled by user");
        }
    }

    private void sendSOS() {
        Log.d(TAG, "SOS triggered from background!");
        Intent intent = new Intent(this, MainActivity.class);
        intent.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK);
        intent.putExtra("auto_sos", true);
        startActivity(intent);
    }

    private void playAlarm() {
        try {
            Uri alarmUri = RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);

```

```

        if (alarmUri == null) {
            alarmUri =
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
        }
        alarmPlayer = MediaPlayer.create(this, alarmUri);
        if (alarmPlayer != null) {
            alarmPlayer.setLooping(true);
            alarmPlayer.start();
        }
    } catch (Exception e) {
        Log.e(TAG, "Error playing alarm", e);
    }
}

private void stopAlarm() {
    if (alarmPlayer != null) {
        try {
            alarmPlayer.stop();
            alarmPlayer.release();
        } catch (Exception e) {
            Log.e(TAG, "Error stopping alarm", e);
        } finally {
            alarmPlayer = null;
        }
    }
}

private void vibratePattern() {
    if (vibrator != null && vibrator.hasVibrator()) {
        long[] pattern = {0, 500, 500, 500, 500, 500}; // vibrate 5 times
        vibrator.vibrate(pattern, -1);
    }
}

private void wakeUpScreen() {
    PowerManager powerManager = (PowerManager)
getSystemService(POWER_SERVICE);
    if (powerManager != null) {
        boolean isScreenOn = powerManager.isInteractive();
        if (!isScreenOn) {
            PowerManager.WakeLock wakeLock = powerManager.newWakeLock(
                PowerManager.SCREEN_BRIGHT_WAKE_LOCK |
PowerManager.ACQUIRE_CAUSES_WAKEUP,
                "Shyra:SOSWakeLock");
            wakeLock.acquire(10000); // keep screen on for 10 seconds
        }
    }
}

```

```

    }
}

private void showSOSNotification(String message) {
    Intent intent = new Intent(this, PowerButtonSOSService.class);
    intent.setAction("CANCEL_SOS");
    PendingIntent pendingIntent = PendingIntent.getService(this, 0, intent,
        PendingIntent.FLAG_UPDATE_CURRENT | PendingIntent.FLAG_IMMUTABLE);

    Notification notification = new NotificationCompat.Builder(this, "SOS_CHANNEL")
        .setContentTitle("SOS Countdown")
        .setContentText(message)
        .setSmallIcon(android.R.drawable.ic_dialog_alert)
        .setPriority(NotificationCompat.PRIORITY_HIGH)
        .setCategory(NotificationCompat.CATEGORY_ALARM)
        .setVisibility(NotificationCompat.VISIBILITY_PUBLIC)
        .addAction(android.R.drawable.ic_menu_close_clear_cancel, "Cancel",
pendingIntent)
        .build();

    startForeground(2, notification);
}

private void showNotification(String message, boolean autoCancel) {
    Notification notification = new NotificationCompat.Builder(this, "SOS_CHANNEL")
        .setContentTitle("Shyra SOS")
        .setContentText(message)
        .setSmallIcon(android.R.drawable.ic_dialog_info)
        .setAutoCancel(autoCancel)
        .build();
    NotificationManager manager = getSystemService(NotificationManager.class);
    manager.notify(3, notification);
}

@Override
public int onStartCommand(Intent intent, int flags, int startId) {
    if (intent != null && "CANCEL_SOS".equals(intent.getAction())) {
        cancelSOSCountdown();
    }
    return super.onStartCommand(intent, flags, startId);
}

@Override
public void onInterrupt() {
    Log.d(TAG, "Service interrupted");
}

```

```

@Override
public void onDestroy() {
    super.onDestroy();
    handler.removeCallbacksAndMessages(null);
    stopAlarm();
}

private void createNotificationChannel() {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        NotificationChannel channel = new NotificationChannel(
            "SOS_CHANNEL",
            "SOS Notifications",
            NotificationManager.IMPORTANCE_HIGH
        );
        channel.setDescription("Notifications for SOS countdown and alerts");
        channel.setSound(null, null);
        channel.enableVibration(false);
        NotificationManager manager = getSystemService(NotificationManager.class);
        manager.createNotificationChannel(channel);
    }
}

private void startForegroundServiceWithNotification() {
    Notification notification = new NotificationCompat.Builder(this, "SOS_CHANNEL")
        .setContentTitle("Shyra SOS")
        .setContentText("Power button SOS is active")
        .setSmallIcon(android.R.drawable.ic_dialog_info)
        .setPriority(NotificationCompat.PRIORITY_LOW)
        .build();
    startForeground(1, notification);
}
}

```

CHAPTER-6

TESTING AND EVALUATION

6.1 Objectives of Testing

The primary objective of testing the Shyra Women Safety Application was to ensure that all implemented features function correctly and reliably under different conditions. Testing was conducted to verify that the system meets the functional and non-functional requirements defined during the design phase.

The testing process aimed to confirm that the application can successfully perform tasks such as user authentication, emergency contact management, sending SOS alerts, retrieving GPS location, capturing media, and triggering emergency alerts through the power button. Another important objective was to ensure that the application provides a smooth and user-friendly experience while maintaining security and reliability.

Testing also focused on identifying errors, bugs, or performance issues in the system so that they could be corrected before the final deployment of the application.

6.2 Types of Testing Conducted

Several types of testing were performed to verify the functionality and reliability of the application.

1. Unit Testing: Individual modules such as login authentication, contact management, location retrieval, and SOS messaging were tested separately to ensure that each module works correctly.

2. Integration Testing: After individual modules were developed, they were combined and tested together to ensure proper communication between different components of the system.

3. Functional Testing: Functional testing was performed to verify that all features operate according to the system requirements. For example, pressing the SOS button should correctly send the emergency message with the location link.

4. Usability Testing: The user interface was tested to ensure that users can easily navigate the application and access safety features without confusion.

5. Performance Testing: The system was tested to check the response time of important operations such as location retrieval and sending SOS alerts.

6. Device Compatibility Testing: The application was tested on different Android devices to ensure that it works correctly across multiple screen sizes and Android versions.

6.3 Testing Environment Setup

The testing environment consisted of both hardware and software components required to run and evaluate the application.

Hardware Environment

- Android smartphone with GPS capability
- Development computer with sufficient RAM and storage

Software Environment

- Android Studio development environment
- Android SDK (API Level 21 and above)
- Firebase Authentication and Realtime Database
- Google Play Services for location services

The application was installed on Android devices and tested under different conditions such as enabling and disabling GPS, testing SMS alerts, and checking media capture functionality.

6.4 Test Scenarios and Outcomes

Test Scenario	Expected Outcome	Actual Outcome
User registration	User account created successfully	Account created successfully
User login	User logs in with valid credentials	Login successful
Add emergency contact	Contact stored in database	Contact saved successfully
Press SOS button	SMS with location sent to contacts	SMS sent with correct location
Share GPS location	Location link generated and shared	Location shared successfully
Capture photo	Camera opens and captures image	Photo captured successfully
Record audio	Audio recording saved and shared	Audio recorded successfully
Power button SOS	Emergency alert triggered	SOS alert triggered correctly

1.5 Test Scenario and Outcomes

6.5 Observed Results vs. Expected Results

During testing, the application behaviour was compared with the expected results defined in the system requirements. In most cases, the application performed exactly as expected.

The SOS alert feature successfully retrieved the user's location and sent SMS messages to emergency contacts. Media capture features worked correctly by allowing users to take photos and record audio files. The power button SOS trigger was also successfully activated through multiple power button presses.

Minor issues related to permission handling and user interface adjustments were observed during initial testing but were corrected during the debugging process.

6.6 Error Handling and Bug Fixes

During the testing phase, several minor issues were identified and resolved to improve the performance and reliability of the application.

Some of the common issues included incorrect permission handling, delays in GPS location retrieval, and minor user interface alignment problems. These issues were fixed by updating the permission management system, optimizing location service requests, and refining layout designs.

Proper error handling mechanisms were also implemented to display appropriate messages when certain features fail, such as when GPS is disabled or when network connectivity is unavailable.

6.7 Testing Tools and Techniques Used

The following tools and techniques were used during the testing process:

- Android Studio Emulator and Real Devices for testing application functionality
- Logcat Debugging Tool for monitoring system logs and identifying errors
- Manual Testing for verifying application features and user interface behavior
- Firebase Console for monitoring database operations and authentication activity

These tools helped identify issues quickly and ensured that the system operates correctly.

6.8 Limitations Observed During Testing

Although the application performed well during testing, certain limitations were observed.

The accuracy of location tracking depends on the availability of GPS signals and device hardware capabilities. In some cases, location retrieval may take longer if GPS signals are weak. The SMS alert system also depends on the availability of mobile network services.

Additionally, the power button SOS feature may behave differently on some devices due to manufacturer restrictions on accessibility services.

Despite these limitations, the overall performance of the application remained stable and reliable.

6.9 Final Evaluation

The final evaluation of the Shyra Women Safety Application shows that the system successfully meets the objectives defined at the beginning of the project. The application provides a comprehensive set of safety features including emergency alerts, live location sharing, media capture, and contact management.

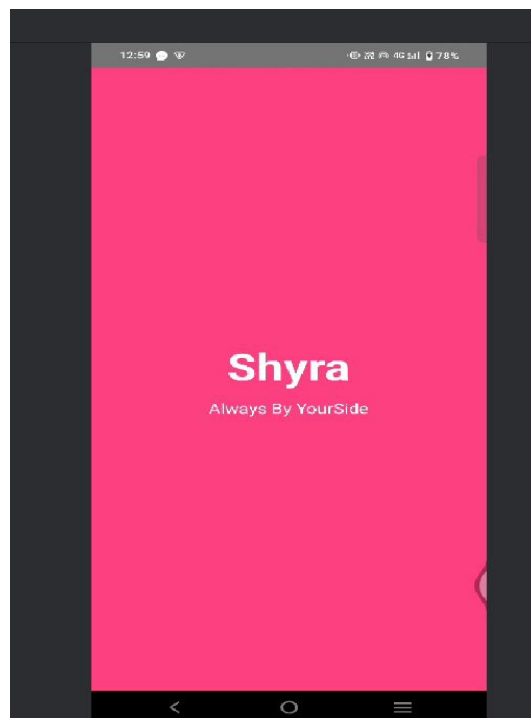
Testing results confirm that the application works efficiently and provides quick response during emergency situations. The system also offers a simple and user-friendly interface, making it easy for users to operate even under stressful conditions.

Overall, the project demonstrates how modern smartphone technology can be effectively used to enhance personal safety. The Shyra application provides a practical and reliable solution that can help users quickly seek assistance during emergencies while maintaining privacy and security.

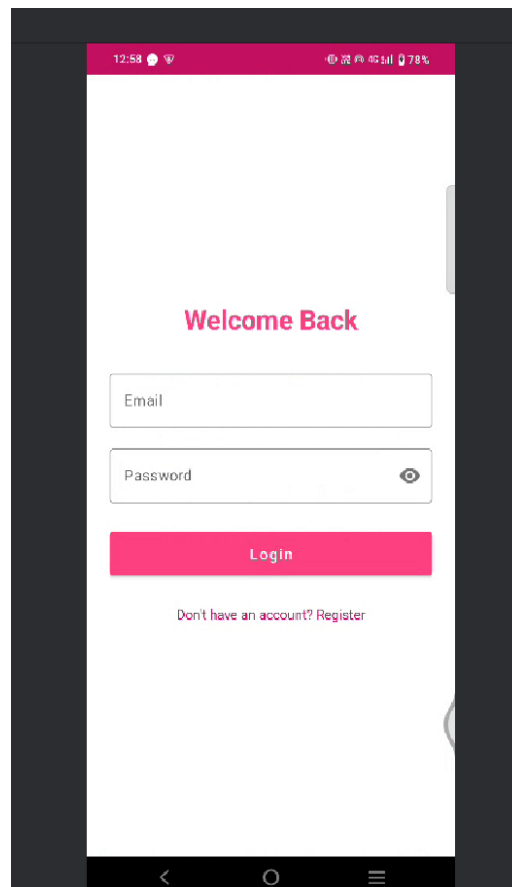
CHAPTER-7

RESULTS

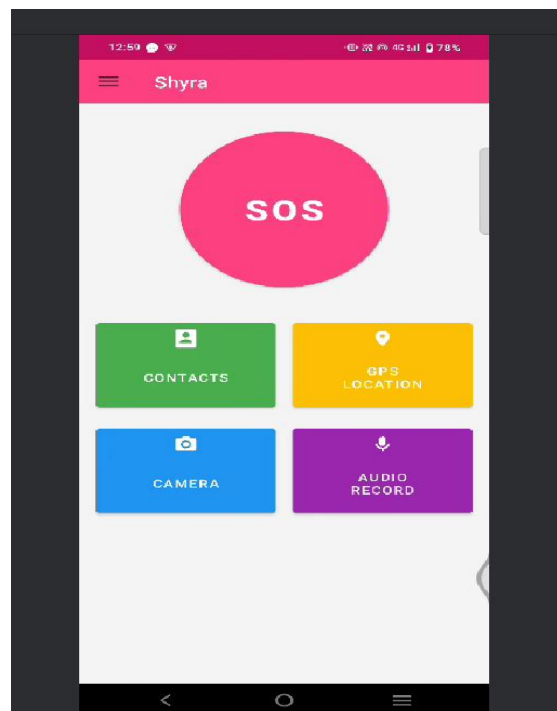
RESULTS



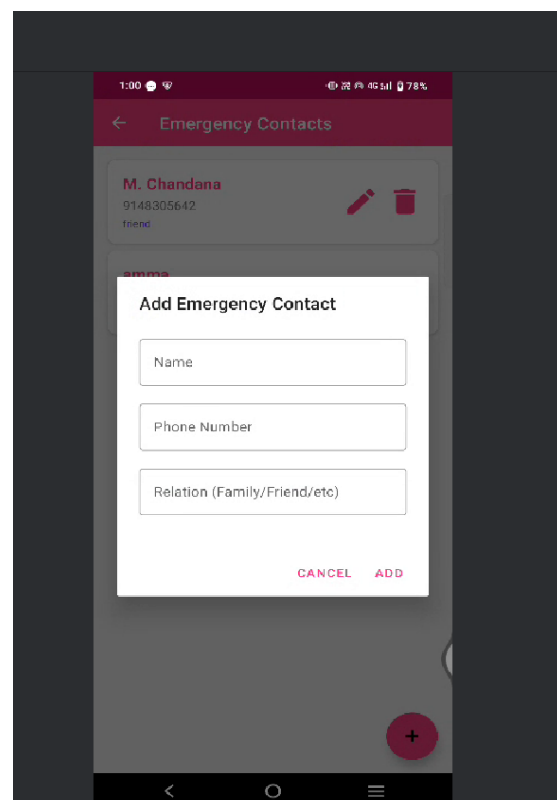
LOGIN PAGE



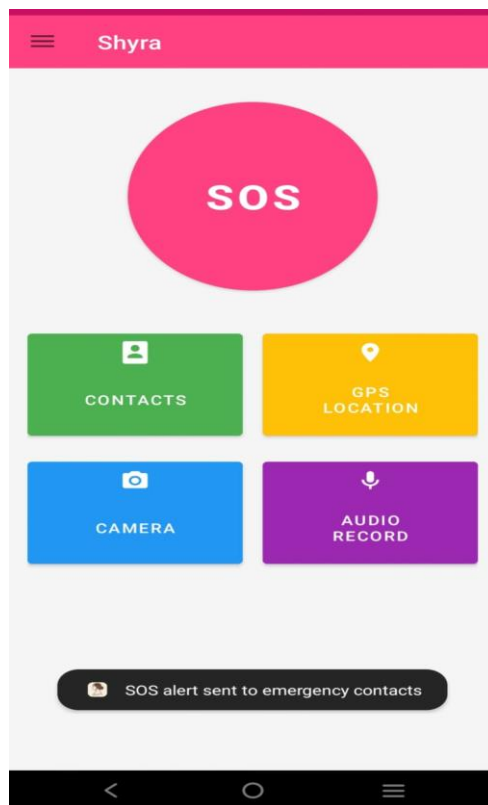
HOME PAGE



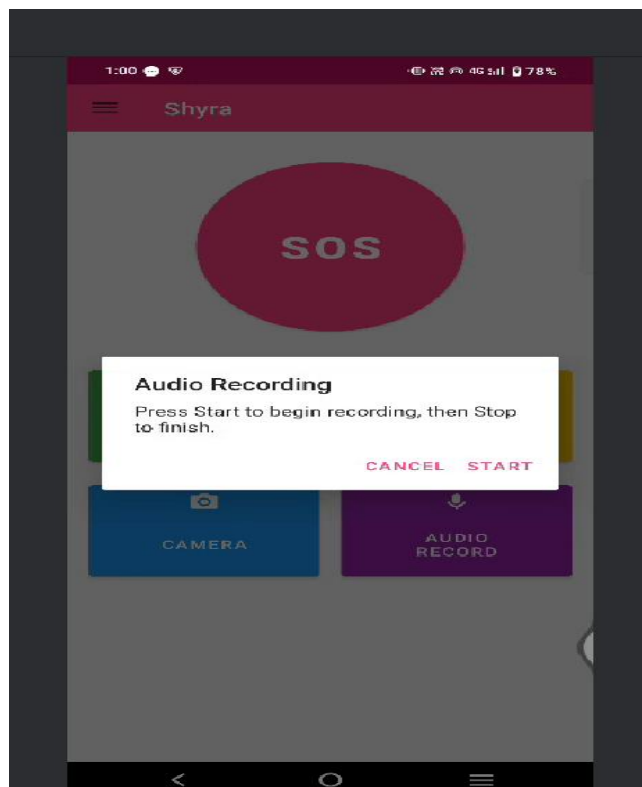
ADD CONTACTS



SOS BUTTON



AUDIO RECORDING



CHAPTER-8

CONCLUSION

8.1 CONCLUSION

The Shyra Women Safety Application was developed with the objective of providing a reliable and easy-to-use mobile safety solution for women during emergency situations. With the increasing concerns about personal safety, especially for women in public spaces, there is a strong need for technological solutions that enable quick communication and rapid response during critical situations.

The application successfully integrates several important safety features into a single Android platform. These features include secure user authentication, emergency contact management, SOS alert messaging with live GPS location, camera photo capture, audio recording, and a power button triggered emergency alert system. These functionalities allow users to quickly notify trusted contacts and share their location when they feel unsafe.

The system was developed using Java programming language, XML-based user interface design, and Firebase services for authentication and database management. Android device services such as GPS, SMS, camera, and microphone were integrated to provide the necessary safety functionalities. The use of Material Design principles ensured that the application interface is simple, user-friendly, and easy to navigate.

The application was tested on Android devices to verify the functionality of each module. Testing results showed that the system performs reliably and responds quickly during emergency situations. Features such as SOS alerts, location sharing, and media capture worked as expected. Minor issues identified during testing were corrected through debugging and refinement.

Overall, the Shyra Women Safety Application demonstrates how modern mobile technology can be effectively used to improve personal safety. The application provides users with a practical tool to quickly communicate with trusted contacts and share critical information during emergencies. By combining multiple safety features into one platform, the system offers a comprehensive and accessible solution for women's safety.

8.2 FUTURE ENHANCEMENTS

Although the Shyra application provides several useful safety features, there are opportunities to further enhance the system in future versions.

One possible improvement is the implementation of a shake-to-SOS feature, where shaking the phone automatically triggers an emergency alert. This would provide another quick and discreet way to activate the SOS system.

Another enhancement could involve wearable device integration, such as connecting the application with smartwatches or fitness bands. This would allow users to send emergency alerts directly from wearable devices without accessing their phones.

The application could also be improved by adding a fake call feature, which can simulate an incoming call to help users escape uncomfortable or dangerous situations. Additionally, a community-based alert system could be introduced, where nearby users or local authorities receive alerts during emergencies.

Future versions of the application may also include geofencing technology, which allows users to define safe and unsafe zones. When the user enters a potentially unsafe area, the system could automatically notify trusted contacts.

Another potential improvement is the integration of AI-based threat detection or voice-activated emergency commands, which could automatically trigger alerts when the system detects signs of distress.

These enhancements would further strengthen the capabilities of the Shyra Women Safety Application and expand its usefulness as a comprehensive personal safety tool.

8.3 REFERENCES

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